

```
import pandas as pd
```

```
df = pd.read_csv(r"C:\Users\carzh\Downloads\archive (1)\messy_ecommerce_sales_data.csv")
df
```

[Show hidden output](#)

```
pd.set_option("display.max.rows", 105)
df
```

[Show hidden output](#)

```
#inspection of the first 10 rows
df.head(10)
```

[Show hidden output](#)

```
#information on our types and values
df.info()
```

[Show hidden output](#)

```
# inspecting last five
df.tail()
```

[Show hidden output](#)

```
#removing any duplicate values
df= df.drop_duplicates()
```

```
#snake caseing the column names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
df.head(5)
```

[Show hidden output](#)

```
print(df.columns.tolist())
```

```
['id', 'customer_name', 'order_id', 'order_date', 'product', 'category', 'quantity', 'price', 'payment_method', 'status', 'total']
```

```
# changing the string date to a numeric date
df.loc[14, "order_date"] = "1/5/2023"
```

```
#change date from string to datetime
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
```

```
#standarizing the product column
df["product"] = df["product"].str.strip().str.lower()
df["product"]
```

[Show hidden output](#)

```
#standarizing the category column
df["category"] = df["category"].str.strip().str.lower()
df['category']
```

[Show hidden output](#)

```
df['category'].unique()
```

```
<ArrowStringArray>
['home', 'electronics', 'sports', 'books', 'clothing', 'electronic', nan]
Length: 7, dtype: str
```

```
df['quantity'].unique()
```

```
<ArrowStringArray>
['3', '2', '1', '5', nan, '4']
Length: 6, dtype: str
```

```
df['price'].unique()
```

Show hidden output

```
df['total'].unique()
```

Show hidden output

```
df['status'].unique()
```

Show hidden output

```
df["category"] = ( df["category"].replace({ "electronic": "electronics"}))
df["category"].unique()
```

```
<ArrowStringArray>
['home', 'electronics', 'sports', 'books', 'clothing', nan]
Length: 6, dtype: str
```

```
#find where category is null by products
df[df['category'].isna()][['product', 'category']]
```

```
product category
```

```
category_map = {
    'biography': 'books',
    'headphones': 'electronics',
    'laptop': 'electronics',
    'smartphone': 'electronics',
    'shoes': 'clothing',
    'jeans': 'clothing',
    'vacuum': 'home',
    'basketball': 'sports'
}
```

```
df['category'] = df['category'].fillna(df['product'].map(category_map))
```

```
df
```

Show hidden output

```
df["quantity"] = df['quantity'].replace({
    "-2": "2",
    "-5": "5",
    "4a": "4"
})
```

```
#change type
df["quantity"] = pd.to_numeric(df["quantity"], errors="coerce")
df["quantity"]
```

Show hidden output

```
# Replace text/formatting issues
df['price'] = df['price'].replace({
    'four hundred': '400',
    '300$': '300',
    '-100': '100'
})
```

```
df["price"] = pd.to_numeric(df["price"], errors="coerce")
```

```
#replacing total values with correct numbers
df['total'] = df['total'].replace({
```

```
'-2000': '400',  
'-2957.65': '2957.65'})
```

```
#change type  
df["total"] = pd.to_numeric(df["total"], errors="coerce")
```

```
df['total'] = df['quantity'] * df['price']
```

```
df[df['total'].isnull()]
```

[Show hidden output](#)

```
df['status'].unique()
```

[Show hidden output](#)

```
df
```

[Show hidden output](#)

Start coding or [generate](#) with AI.

```
# Remove records that still cannot be used in sales analysis  
clean_df = df.dropna(  
    subset=["order_date", "category", "quantity", "price", "total"]  
)
```

```
clean_df.info()
```

[Show hidden output](#)

```
clean_df.isna().sum()
```

[Show hidden output](#)

```
# Save cleaned data and export to sql  
clean_df.to_csv("cleaned_ecommerce_sales_data.csv", index=False)
```

```
%pip install sqlalchemy pymysql
```

[Show hidden output](#)

```
import pandas as pd  
from sqlalchemy import create_engine, text
```

```
clean_df.to_sql(  
    name="ecommerce_sales",  
    con=engine,  
    if_exists="replace",  
    index=False  
)
```

```
90
```

```
query = "SELECT * FROM ecommerce_sales LIMIT 5"  
  
uploaded_data = pd.read_sql(query, engine)  
  
uploaded_data
```

	id	customer_name	order_id	order_date	product	category	quantity	price	payment_method	status	total
0	100	Customer_100	ORD-41285	2024-11-22	blender	home	3.0	38.00	Cash on Delivery	Shipped	114.00
1	102	Customer_102	ORD-84355	2024-12-23	tennis racket	sports	1.0	389.05	PayPal	Delivered	389.05
2	103	Customer_103	ORD-57811	2025-03-19	science	books	5.0	233.92	PayPal	Processing	1169.60
3	104	Customer_104	ORD-93614	2025-10-20	biography	books	1.0	552.51	Cash on Delivery	Processing	552.51
4	105	Customer_105	ORD-22442	2024-11-20	tennis racket	sports	3.0	122.06	Cash on Delivery	Cancelled	366.18

pip install -U notebook-as-pdf

Collecting notebook-as-pdf
 Downloading notebook_as_pdf-0.5.0-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: nbconvert in .\anaconda3\Lib\site-packages (from notebook-as-pdf) (7.17.1)
Collecting pypeteer (from notebook-as-pdf)
 Downloading pypeteer-2.0.0-py3-none-any.whl.metadata (7.1 kB)
Collecting PyPDF2 (from notebook-as-pdf)
 Downloading pypdf2-3.0.1-py3-none-any.whl.metadata (6.8 kB)
Requirement already satisfied: beautifulsoup4 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (4.15.0)
Requirement already satisfied: bleach!=5.0.0 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (5.9.1)
Requirement already satisfied: defusedxml in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (0.7.1)
Requirement already satisfied: Jinja2>=3.0 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (3.1.6)
Requirement already satisfied: jupyter-core>=4.7 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (5.9.1)
Requirement already satisfied: jupyterlab-pygments in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (0.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (3.0.3)
Requirement already satisfied: mistune<4,>=2.0.3 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (3.3.3)
Requirement already satisfied: nbclient>=0.5.0 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (0.11.0)
Requirement already satisfied: nbformat>=5.7 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (5.10.4)
Requirement already satisfied: packaging in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (26.0)
Requirement already satisfied: pandocfilters>=1.4.1 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (1.5.1)
Requirement already satisfied: Pygments>=2.4.1 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (2.20.0)
Requirement already satisfied: traitlets>=5.1 in .\anaconda3\Lib\site-packages (from nbconvert->notebook-as-pdf) (5.15.0)
Requirement already satisfied: webencodings in .\anaconda3\Lib\site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->notebook-as-pdf) (0.11.0)
Requirement already satisfied: tinycss2>=1.1.0 in .\anaconda3\Lib\site-packages (from bleach[css]!=5.0.0->nbconvert->notebook-as-pdf) (1.1.0)
Requirement already satisfied: platformdirs>=2.5 in .\anaconda3\Lib\site-packages (from jupyter-core>=4.7->nbconvert->notebook-as-pdf) (4.3.6)
Requirement already satisfied: jupyter-client>=7.0.0 in .\anaconda3\Lib\site-packages (from nbclient>=0.5.0->nbconvert->notebook-as-pdf) (7.1.0)
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\Lib\site-packages (from jupyter-client>=7.0.0->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (2.8.2)
Requirement already satisfied: pyzmq>=25.0 in .\anaconda3\Lib\site-packages (from jupyter-client>=7.0.0->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (25.1.2)
Requirement already satisfied: tornado>=6.4.1 in .\anaconda3\Lib\site-packages (from jupyter-client>=7.0.0->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (6.4.1)
Requirement already satisfied: typing-extensions>=4.13.0 in .\anaconda3\Lib\site-packages (from jupyter-client>=7.0.0->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (4.13.0)
Requirement already satisfied: fastjsonschema>=2.15 in .\anaconda3\Lib\site-packages (from nbformat>=5.7->nbconvert->notebook-as-pdf) (2.20.1)
Requirement already satisfied: jsonschema>=2.6 in .\anaconda3\Lib\site-packages (from nbformat>=5.7->nbconvert->notebook-as-pdf) (4.23.0)
Requirement already satisfied: attrs>=22.2.0 in .\anaconda3\Lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (25.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in .\anaconda3\Lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (2023.12.1)
Requirement already satisfied: referencing>=0.28.4 in .\anaconda3\Lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (0.35.1)
Requirement already satisfied: rpds-py>=0.7.1 in .\anaconda3\Lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (0.20.1)
Requirement already satisfied: six>=1.5 in .\anaconda3\Lib\site-packages (from python-dateutil>=2.8.2->jupyter-client>=7.0.0->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (1.16.0)
Requirement already satisfied: soupsieve>=1.6.1 in .\anaconda3\Lib\site-packages (from beautifulsoup4->nbconvert->notebook-as-pdf) (2.6.0)
Collecting appdirs<2.0.0,>=1.4.3 (from pypeteer->notebook-as-pdf)
 Downloading appdirs-1.4.4-py2.py3-none-any.whl.metadata (9.0 kB)
Requirement already satisfied: certifi>=2023 in .\anaconda3\Lib\site-packages (from pypeteer->notebook-as-pdf) (2026.7.22)
Requirement already satisfied: importlib-metadata>=1.4 in .\anaconda3\Lib\site-packages (from pypeteer->notebook-as-pdf) (8.7.0)
Collecting pyee<12.0.0,>=11.0.0 (from pypeteer->notebook-as-pdf)
 Downloading pyee-11.1.1-py3-none-any.whl.metadata (2.8 kB)
Requirement already satisfied: tqdm<5.0.0,>=4.42.1 in .\anaconda3\Lib\site-packages (from pypeteer->notebook-as-pdf) (4.68.2)
Collecting urllib3<2.0.0,>=1.25.8 (from pypeteer->notebook-as-pdf)
 Downloading urllib3-1.26.20-py2.py3-none-any.whl.metadata (50 kB)
Collecting websockets<11.0,>=10.0 (from pypeteer->notebook-as-pdf)
 Downloading websockets-10.4.tar.gz (84 kB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Requirement already satisfied: colorama in .\anaconda3\Lib\site-packages (from tqdm<5.0.0,>=4.42.1->pypeteer->notebook-as-pdf) (0.4.6)
Requirement already satisfied: zipp>=3.20 in .\anaconda3\Lib\site-packages (from importlib-metadata>=1.4->pypeteer->notebook-as-pdf) (3.20.2)
Downloading notebook_as_pdf-0.5.0-py3-none-any.whl (6.5 kB)

Start coding or [generate](#) with AI.

